Storage Format

key (string)	value (string)
""	{"req1": "{}", "req2": "{"area1": "{}"}" "req3": "{"areaA": "{}", "areaB": "{}"}", "req4": "{"area_1": "{"course_1": "not satisfied"}" }" "req5": "{"area_A": "{"course_A": "satisfied", "course_B": "not satisfied", "course_C": "in progress"}", "area_B": "{"course_A: "in progress", "course_B": "satisfied", "course_C": "not satisfied"}"}" }

Why Use map?

- readable iteration (no Object.keys (localstorage...)) → returns {'0', '1', '2', ...}
  just [key, value], myMap.keys(), or myMap.values()
- Can get size of data through .size property if needed
- readable setting key value pairs (myMap.set('key', 'value') vs (myObj.key = 'value'))

```
getMapOfWebData()
    {"req1" => "{}",
     "req2" => "{"area1": "{}"}"
     "req3" => "{"areaA": "{}",
                "areaB": "{}"}",
     "req4" => "{"area_1":
       "{"course_1": "not satisfied"}"
       }"
     "req5" => "{"area_A": "{"course_A":
      "satisfied", "course_B": "not satisfied",
      "course_C": "in progress"}",
      "area_B": "{"course_A: "in progress",
      "course_B": "satisfied", "course_C":
      "not satisfied"}"}"
      }
```

key: String [requirement]

value: String [requirement data]

- areas
  - courses
    - status

getMapOfRequirementData('req5')

```
{"area_A" => "{"course_A": "satisfied",
                "course_B": "not satisfied",
                "course_C": "in progress"
               }",

"area_B" => "{"course_A: "in progress",
              "course_B": "satisfied",
              "course_C": "not satisfied"
              }"
}
```

Key: String [area]

Value: String [area data]

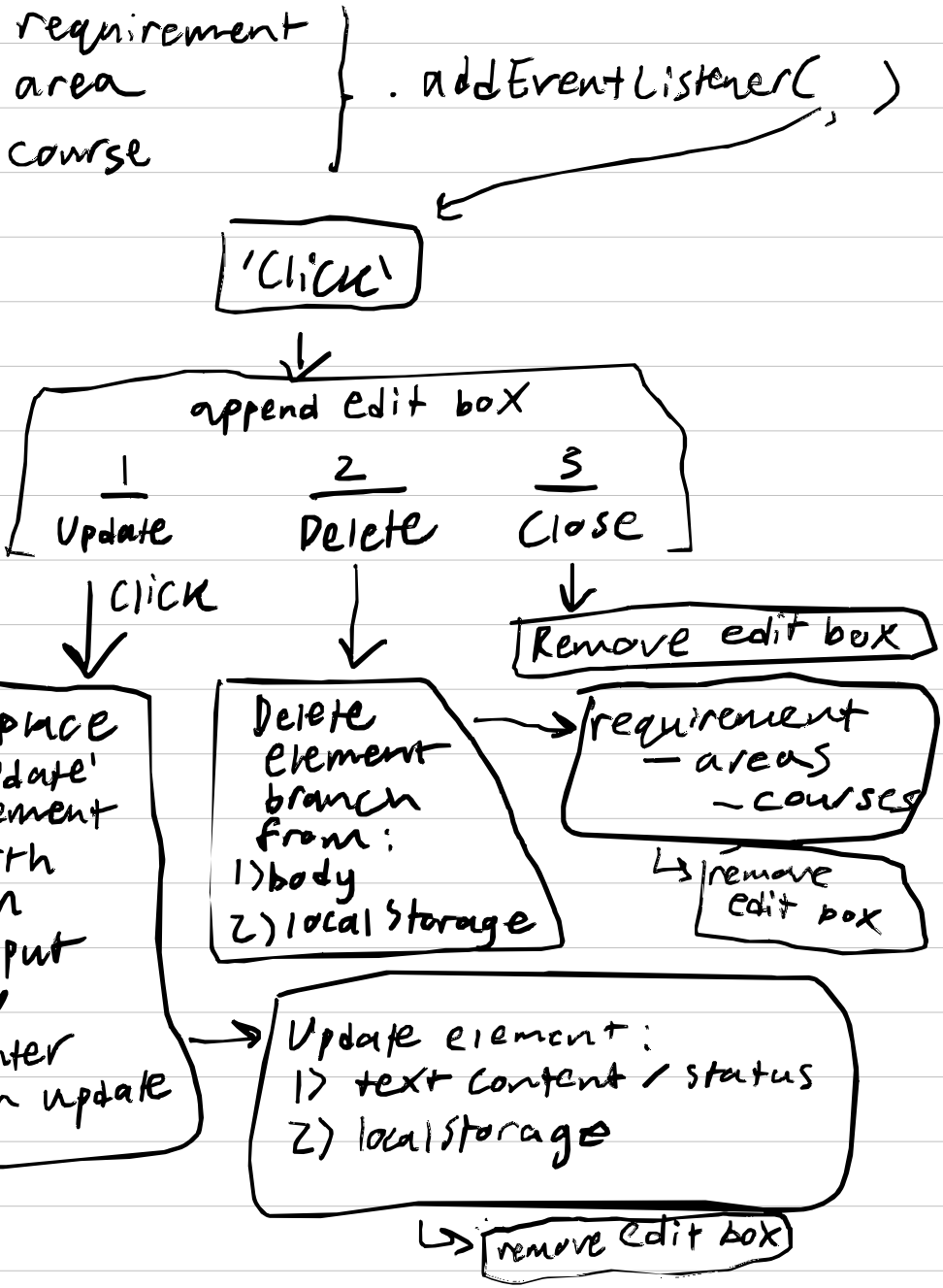
- courses
  - status

getMapOfAreaData('req5', 'course_B')

{"course_A ⇒ "in progress",
"course_B" ⇒ "satisfied",
"course_C" ⇒ "not satisfied"
}

key : string [course]
value : string [course data]
- status

Editing Elements



Deleting From localStorage

key (string)	value (string)
""	{"req1": {}, "req2": {"area1": "{}"} "req3": {"areaA": "{}", "areaB": "{}"}, "req4": {"area_1": {"course_1": "not satisfied"} } "req5": {"area_A": {"course_A": "satisfied", "course_B": "not satisfied", "course_C": "in progress"}, "area_B": {"course_A: "in progress", "course_B": "satisfied", "course_C": "not satisfied"} } }

Deleting From localStorage cont.

Deleting requirements:

1) json = getJSONOfWebData() → {...}
2) delete json[deleted_requirement_string]

Deleting areas:

1) (see req)
2) delete json[requirement_string]
[deleted_area_string]

Deleting courses:

1) (see req)
2) delete json[requirement_string]
[area_string]
[deleted_course_string]

Updating localStorage

Updating requirements:

1) json = getJSONOfWebData() → {...}
2) updated_json = {}
3) for (const [req_str, data] of Object.entries(json))
   {
      if (req_str !== update_str){
         updated_json[req_str] = data;
      }
      else if (req_str === update_str){
         updated_json[new_str] = data;
      }
   }
4) stringified_update_json
      = JSON.stringify(updated_json);
5) localStorage.setItem('',
      stringified_updated_json);

Updating localStorage cont.

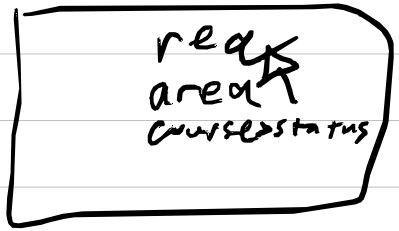
Updating Areas:

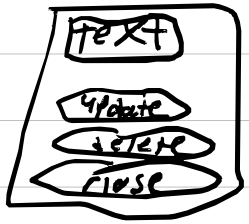
1) web_json = getJSONOfWebData() → {...}
2) requirement_json = web_json[requirement_str]
3) update_req_json = {}
4) for(const [area_str, data] of
   Object.entries(requirement_json)){
   if(area_str !== update_area_str){
   update_req_json[area_str] = data;
   }
   else if(area_str === update_area_str)
   {
   update_req_json[new_str] = data;
   }
   }
5) web_json[requirement_str] = update_req_json;
6) stringified_update_json
   = JSON.stringify(web_json)
7) localStorage.setItem(`',
   stringified_updated_json);

Updating localStorage cont.

Updating courses:

```
1) web_json = getJSONOfWebData() → {...}
2) req_json = web_json[req_str];
3) area_json = req_json[area_str];
4) update_area_json = {};
5) for(const [course_str, data] of
     Object.entries(area_json)){
         if(course_str !== update_course_str){
             update_area_json[course_str] = data;
         }
         else if(course_str===update_course_str)
         {
             update_area_json[new_str]
             = data;
         }
     }
6) req_json[area_str] = update_area_json;
7) web_json[req_str] = req_json;
8) stringified_update_json
     = JSON.stringify(web_json)
9) localStorage.setItem('',
     stringified_updated_json);
```





Parsing requirement update

1) Click on req_text
2) Click on update_option
3) Parse update_input to local storage
4) change text content of current requirement

① Upon clicking requirement:

1) append edit box to rectangle
2) set update_req_string = event.target.textContent

② Upon Inputting Update:

1) Parse update_req_string and old data to local storage
2) set textContent of requirement to new_req_string
3) remove edit box